

DisasTRACE Overall System Documentation

Current system, feature, security, and operational reference

Prepared 11 September 2026. This document describes the current implemented repository state and should be regenerated when material behavior changes.

1. Executive Summary

DisasTRACE is a centralized digital emergency incident reporting and ambulance dispatch platform for CDRRMO Baliwag City and the Public Assistance and Command Center (PACC). It connects residents, guests, ambulance responders, PACC dispatchers, and CDRRMO Super Administrators through an Android Expo mobile app and a Next.js command dashboard.

Operational principle

The server and database are the source of truth for identity, verification, incident state, dispatch ownership, expiry, location integrity, and guest access. Mobile and dashboard clients present and reconcile that state.

Current outcomes

- Residents and responders are blocked until CDRRMO Super Admin approval.
- Guests can reach an emergency chatbot without creating an account, with scoped status access only.
- High-confidence emergencies enter FIFO ambulance dispatch; uncertain, non-emergency, suspicious, duplicate, or unassigned cases remain visible to PACC.
- PACC and CDRRMO can monitor multiple active ambulances, incident severity, responder presence, and road routes in real time.
- Responders can complete PCR and Driver's Trip Ticket forms, capture electronic signatures, save drafts offline, and submit when connected.

2. Users and Responsibilities

Public User / Resident

Registers with personal details, government ID, and OTP-verified Philippine phone number. After approval, the resident uses the chatbot or legacy intake, provides foreground GPS, tracks an assigned response, receives report updates, and views report history and hospitals.

Guest Reporter

Uses Login as Guest during an emergency. No account, tabs, history, profile, or protected API access is granted. The guest supplies a callback number, automatic GPS, nearby landmark, incident details, affected-person count, condition, and required evidence, then receives a random report access token for scoped status refresh.

Ambulance Responder

After approval and one-device sign-in, the responder manages duty status, receives dispatch offers, accepts and navigates to incidents, confirms arrival, completes operational reports, and synchronizes offline drafts. Foreground location is required for app use; background tracking is an on-duty capability.

PACC Admin / Dispatcher

Triages incoming reports, monitors the live map and responder roster, records coordination with partner agencies, dispatches or overrides responder selection, and handles cases requiring human review.

CDRRMO Super Admin

Owns registration approval/rejection, user management, bans and device release, analytics, exports, security audit review, system settings, and operational oversight. PACC Admins do not approve registrations.

3. System Architecture

Technology stack

- Next.js 16.2.6, React 19, and TypeScript for the authenticated web dashboard and REST API route handlers.
- Expo / React Native for the Android resident and responder mobile clients.
- Supabase Auth for identity, sessions, JWT claims, and browser/mobile authentication.
- Supabase PostgreSQL with Drizzle ORM for relational data, migrations, transactions, and constraints.
- Supabase Realtime for authenticated live incident, dispatch, telemetry, notification, and coordination updates.
- Supabase Storage for IDs, scene evidence, avatars, and exported report assets.
- OpenFreeMap tiles with MapLibre on web and mobile; no map API key is required.
- textbee.dev for OTP SMS and Expo Push Notifications/FCM for responder dispatch offers when Android is backgrounded or stopped.

Data boundaries

The web dashboard owns command and administration workflows. The mobile app consumes validated REST endpoints and uses Realtime where authenticated access is appropriate. Binary assets are stored in Supabase Storage; relational references and evidence metadata remain in PostgreSQL. Guest status uses a token-scoped endpoint rather than anonymous database subscriptions.

Core data

Users, verification requests, incidents, dispatch offers, responder reports, patient care reports, driver trip tickets, notifications, audit logs, status histories, hospitals, support content, system settings, mobile push tokens, mobile device sessions, and guest device quota records are represented in the Drizzle schema.

4. Emergency Intake and Triage

Required intake controls

- Foreground location permission is mandatory and coordinates must resolve inside the official Baliwag City barangay polygon set.
- The server derives the official barangay label from GPS; user-entered landmarks are references, not authoritative incident locations.
- Guest evidence is uploaded through a protected server endpoint with 5 MB JPEG, PNG, or WebP validation; unauthenticated clients do not receive direct Storage write access.
- The same-submission retry is idempotent for the same phone and device digest.

Deterministic initial triage

The system records one of four reasons: `HIGH_CONFIDENCE_EMERGENCY`, `HIGH_CONFIDENCE_NON_EMERGENCY`, `UNCERTAIN_INCOMPLETE`, or `SUSPICIOUS_POSSIBLE_PRANK`. Only the first class starts automatic ambulance dispatch. PACC receives the other classes and may override the classification. The chatbot may use reviewed English, Filipino, and Taglish signal rules and a server-only DeepSeek gateway, but it never dispatches on its own.

Duplicate and capacity protection

Likely duplicate reports are held for PACC review using a configurable Haversine radius and recent time window. Guest lifetime limits are enforced per normalized Philippine phone and SHA-256 Android app-scoped device digest; raw device identifiers are never stored. The CDRRMO Super Admin controls the phone limit from the singleton system settings row.

5. Dispatch, Tracking, and Recovery

Dispatch lifecycle

- A responder offer has a countdown and is protected until it expires. Acceptance atomically changes the incident to EN_ROUTE.
- Expiry is driven by the Supabase Cron/pg_net scheduler and is also reconciled by scoped status reads; it does not depend on a responder phone remaining online.
- If no alternate unit is available, the report remains visible as PACC reassignment required. PACC can use guarded Override Dispatch for a fresh eligible responder.
- Automatic dispatch, manual dispatch, PACC rejection, expiry cascading, and acceptance serialize around locked request state and atomic responder reservation.

Tracking

Residents and guests see ambulance movement only after a responder accepts. The web command map supports concurrent active ambulances, severity-labelled incident markers, responder identity, active incident relationships, telemetry subscriptions, and cached road routes. Guest tracking refreshes a token-scoped status endpoint every three seconds; it never reads protected incident tables.

Agency coordination

PACC may record PNP, BFP, CDRRMO, Barangay, DSWD, and Hospital coordination on the verification request. The client derives readable coordination status and appends actual incident and responder state. These agencies are coordination entries; only ambulance responders are auto-dispatched by this system.

6. Responder Documentation

The responder workflow includes a pre-hospital Patient Care Report and Driver's Trip Ticket aligned to the CDRRMO paper forms. Captured details include vitals, GCS, SAMPLE, pain assessment, narrative, handoff and referral information, mileage, fuel, travel date, grease log, passenger records, driver contact, and responder identity.

Patient, witness, accomplished-by, driver, and authorized-passenger signatures are drawn as SVG paths, persisted in drafts and report payloads, rendered in the web detail sheet, and converted to embedded images for PDF export.

Offline behavior

Complete responder form drafts are persisted locally. Intentional Save as Draft is distinct from background recovery. When connectivity returns and drafts remain unsent, native high-priority reminders repeat every five minutes and an in-app banner links to review and submit.

7. Security, Integrity, and Privacy

- Role-based access is enforced through Supabase JWT claims, Next.js proxy checks, API authorization, and database policies.
- Mobile residents and responders are bound to one active application session using only a SHA-256 app-scoped device digest and JWT session ID. A different active device receives a logout-first response; CDRRMO/PACC browser sessions are excluded.
- Signing in revokes other mobile sessions. Sign-out releases the matching binding, and administrators can release a device from User Management.
- Mock Android locations are rejected. Implausible responder movement retains the last trusted position and creates a rate-limited audit event. Evidence-photo GPS is advisory and cannot replace captured report GPS.
- Guest access tokens are scoped to one report. Guest quota records retain hashes and counts, not raw Android identifiers.

Known remaining hardening

Google Play Integrity is not yet configured. Stronger modified/rooted-client resistance requires a Play Console/Google Cloud project, release signing, a request-bound integrity token, and server-side verdict verification.

8. Administration and Reporting

The dashboard provides KPI cards, incident distribution, recent reports, responder roster, live maps, verification queues, reports management, user management, security audit trail, support and FAQs, and a CDRRMO-only analytics dashboard. Analytics supports daily, weekly, and monthly trends, incident type frequency, reported/verified/pending/resolved counts, resolution-time summaries, and data-derived preparedness guidance.

Branded PDF exports are available for incident details, report summaries, users, Patient Care Reports, and Driver's Trip Tickets. Exports include Baliwag CDRRMO branding, dynamic multi-line layout handling, official location labels, and embedded signatures where applicable.

9. Operational Deployment Notes

- Run the registered Drizzle migration sequence, including the mobile session, guest quota, barangay, evidence geotag, and dispatch expiry scheduler migrations.
- Configure Supabase Auth site and redirect URLs, JWT role synchronization, RLS/RBAC SQL, Storage buckets, Realtime publications, and Supabase Cron/pg_net scheduler secrets.
- Configure textbee.dev OTP credentials and Expo Push/FCM credentials for background responder offers.
- Seed official Baliwag barangay boundaries and hospitals, initialize singleton system settings, and verify the service-area resolver.
- Validate production environment variables, HTTPS, app deep links (including disastrace://reset-password), release signing, and database backups before field deployment.

10. Current Scope and Success Criteria

In scope: Android resident/responder clients, Next.js command dashboard, Supabase Auth/RBAC, verification, emergency intake, triage, dispatch, live tracking, Realtime, offline responder forms, PDF exports, analytics, notifications, OTP, and operational auditability. iOS, billing, predictive emergency detection, and non-medical response automation remain out of scope.

The system is successful when approved residents can report with authoritative GPS, Super Admins can verify accounts, dispatchers can triage and assign the nearest eligible responder, responders can accept/navigate/document, reporters can track an accepted response, and administrators can audit, analyze, and export the resulting record.